

Phase 2 — non-self-consistent RPA, peak memory and wall time

Mercury (Z=80), all-electron, one outer cycle. Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). E_c^{RPA} on frozen GGA_PBE orbitals (xc='RPA@DFT'), against self-consistent RPA in the corrected code. No OEP equation is solved and no correlation energy density is built, so dE is physics, not error.

Radial mesh sweep (Lmax = 30)

N_fe	n_quad	RPA@DFT (MiB)	scRPA (MiB)	reduction	RPA@DFT (s)	scRPA (s)	speedup	dE (Ha)
10	450	241	263	-8.5%	6.6	17.8	2.70x	-3.39e+00
15	675	333	375	-11.1%	13.2	45.7	3.46x	-3.39e+00
25	1125	586	650	-9.9%	40.5	149.4	3.69x	-3.39e+00
40	1800	1194	1390	-14.1%	129.3	460.8	3.56x	-3.39e+00

Angular sweep (N_fe = 40)

Lmax	n_quad	RPA@DFT (MiB)	scRPA (MiB)	reduction	RPA@DFT (s)	scRPA (s)	speedup	dE (Ha)
5	1800	916	1138	-19.5%	35.0	107.2	3.06x	-3.39e+00
10	1800	967	1154	-16.2%	54.0	177.7	3.29x	-3.39e+00
20	1800	1094	1258	-13.1%	91.4	314.4	3.44x	-3.39e+00
30	1800	1194	1390	-14.1%	129.3	460.8	3.56x	-3.39e+00

reduction = (RPA@DFT - scRPA)/scRPA on peak memory speedup = scRPA/RPA@DFT on wall time dE = RPA@DFT - scRPA on total energy
Peak memory is the sampled maximum resident set size; kernel ru_maxrss and SLURM MaxRSS agree with it to ~1%.